INCH-POUND

MIL-C-24643/22C
14 March 1994
SUPERSEDING
MIL-C-24643/22B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 5000 VOLTS, TYPE LS5KVTSGU
(INCLUDING VARIATION LS5KVTSGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, (see table I for size).
* Second - Silicone rubber - glass tape insulation (see table I for diameter). Standard identification code applied by 3 or letter identification code by method 5.
* Third - The three conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly. Cabling sequence shall be A, B and C.
Fourth - A binder or combination binder and barrier.

Construction variant for type LS5KVTSGU

* Fifth - Cross-linked polyolefin jacket. Minimum average thickness 0.090 inch.

Construction variant for type LS5KVTSGA

* Fifth - Cross-linked polyolefin jacket. Minimum average thickness 0.090 inch.
* Sixth - Braided metal armor.

AMSC N/A FSC 6145
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* TABLE I. Details.

1/ Military part no. M24643/22	Size 2/	Conductors Size		Type's overall diameter (inches)				Dia over insulation min avg (in.)	Conductor resistance per 1000 ft at 25°C max (ohms)	Insulation resistance (meg-ohms)	Cold bending mandrel dia (max) (in.)	Accelerated service loading ampere
		Navy std	AWG	LS5KVTSGU		LS5KVTSGA						
				min	max	min	max					
-01UN	100		0 (class D)		1.74		1.79	0.675	0.106	400	22	295
-02UN	150		000 (class D)		1.95		2.00	.769	.0674	350	24	400
-03UN	250		250,000 (class C)		2.22		2.27	.889	.0453	300	28	535
-04UN	350		350,000 (class D)		2.45		2.50	.994	.0321	265	30	665
-05UN	400	400 (127)	----		2.60		2.65	1.054	.0273	250	32	750

1/ Unarmored. When armored option is required, see MIL-C-24643 for configuration of part number.

2/ Type LS5KVTSGU and LS5KVTSGA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	13,500
Insulation resistance - megohms/1000 feet, minimum	
* Conductor to conductor	(see table I)
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure

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EXAMINATION AND TESTS (continued):	Requirements
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas Flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	125
* Cable sealant removability	No failure
Armor (type LS5KVTSGA only) - conformance to material, construction and coverage	No failure
*Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:	Requirements
* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:	
Cold working (minus 20 ± 2°C)	No damage
Gas Flame (3 hours)	No failure
Accelerated service - sizes 50 through 800 (see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18

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QUALIFICATION INSPECTION (continued):	Requirements
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: All types and sizes 1000 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
Navy - SH

Review Activities:
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